

Real-time ML-based Defense Against Malicious Payload in Reconfigurable Embedded Systems

Abstract

Motivation:

- ❖ Field-Programmable Gate Arrays (FPGAs) deliver **high performance with lower power consumption**, making them **ideal for real-time, adaptive, and cloud-based systems**. Major cloud providers (e.g., Microsoft, IBM, etc.) integrate FPGAs for tasks like deep-learning inference and accelerated networking.
- ❖ However, their reconfigurable nature introduces new security risks—particularly in multi-tenant cloud environments where shared hardware can be reprogrammed using malicious bitstreams. These can trigger hardware Trojans, enabling **denial-of-service (DoS), power abuse, data leakage, and covert channel attacks**.

Ongoing Issues:

- Bitstream-level attacks can bypass traditional security.
- Vendor-controlled bitstream generation compromises IP privacy.
- Reverse engineering-based Trojan detection is slow and requires netlist/source files.

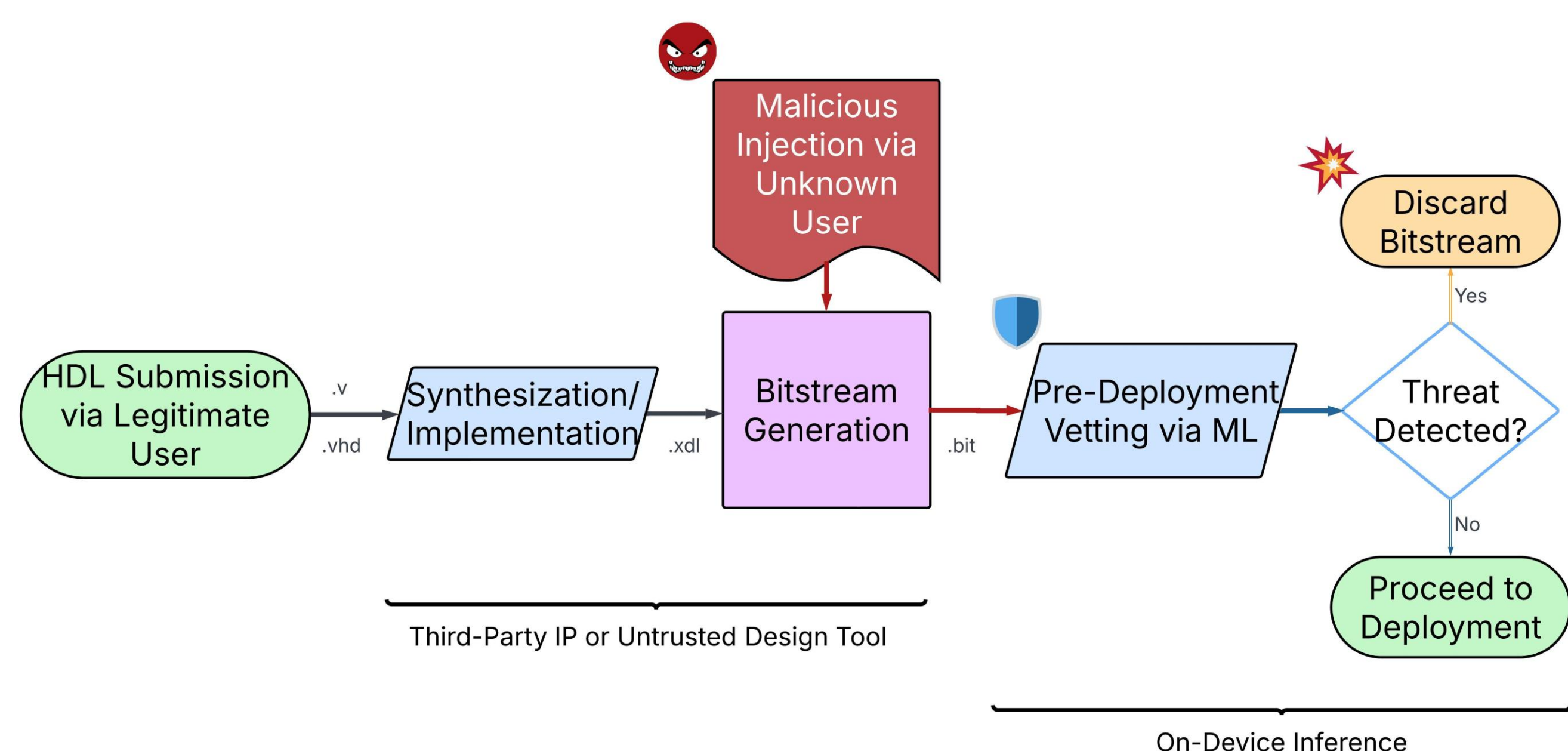


Figure 1. Threat Model for Detecting Trojan-Injected FPGA Bitstreams in Multi-Tenant Systems.

Our Solution:

We propose a **binary-level bitstream analysis approach** which:

- ✓ Detects hardware Trojans **pre-deployment**.
- ✓ Runs efficiently on **lightweight, resource-constrained embedded systems**.
- ✓ Builds on machine learning to classify malicious configurations efficiently.
- ✓ Works in **real-time, without netlist access or backwards analysis**.

Target Platform – Xilinx PYNQ:

- Hybrid architecture (ARM + FPGA) enables on-chip ML inference with **real-time hardware monitoring**.
- Python ecosystem **seamlessly integrates ML libraries** for rapid development and testing.
- Bitstream reconfigurability exposes real FPGA behavior and vulnerabilities.
- Resource constraints reflect the challenges of **real-world IoT and edge deployments**.
- Open-source & accessible; encouraging reproducibility, experimentation, and community collaboration in **security-focused FPGA research**.

Methodology

- ❑ The overall methodology is outlined in **Figures 2 and 3**, which illustrate the end-to-end pipeline used for real-time bitstream-level Trojan detection.
- ❑ Starting with benchmark designs sourced from Trust-Hub, AES-128 and RS232 variants were synthesized into .bit files and labeled as benign, malicious, or empty.
- ❑ Byte-frequency features were extracted from each file, followed by dimensionality reduction using **Truncated Singular Value Decomposition (TSVD)** to retain structural relevance while improving computational efficiency.
- ❑ To address dataset imbalance, the **Synthetic Minority Oversampling Technique (SMOTE)** was applied to oversample minority classes before training.
- ❑ Multiple classifiers were evaluated **using k-Fold Cross Validation (kFCV)**. The best-performing model was serialized and deployed to the PYNQ-Z1, where it performed on-device inference using the same preprocessing pipeline to generate predictions.
- ❑ This final step validated the feasibility of real-time Trojan detection in a constrained embedded environment without reliance on external computation.

Training and Deployment

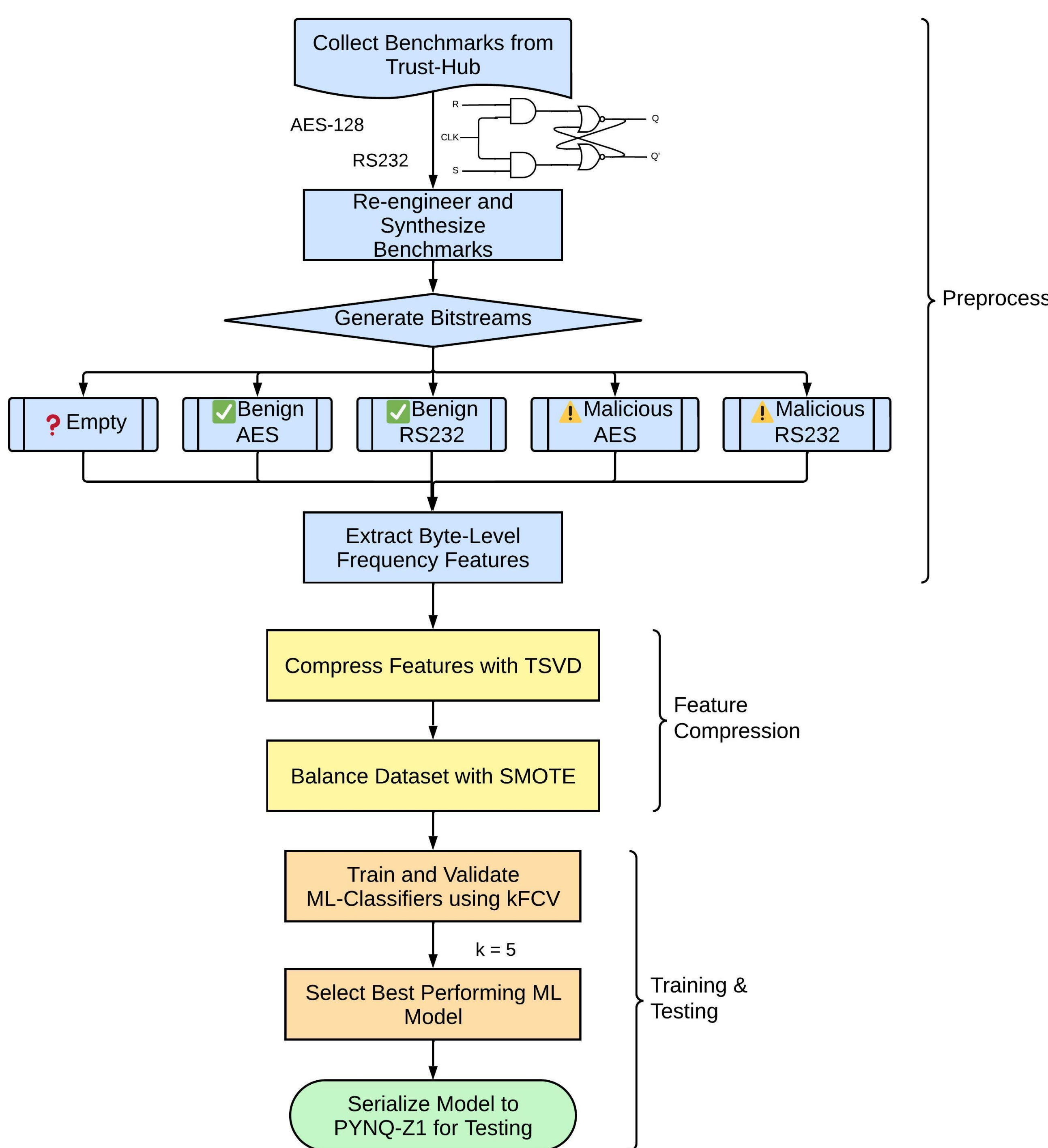


Figure 2. Training and Deployment Pipeline for Bitstream-Level Trojan Detection.

Runtime Prediction

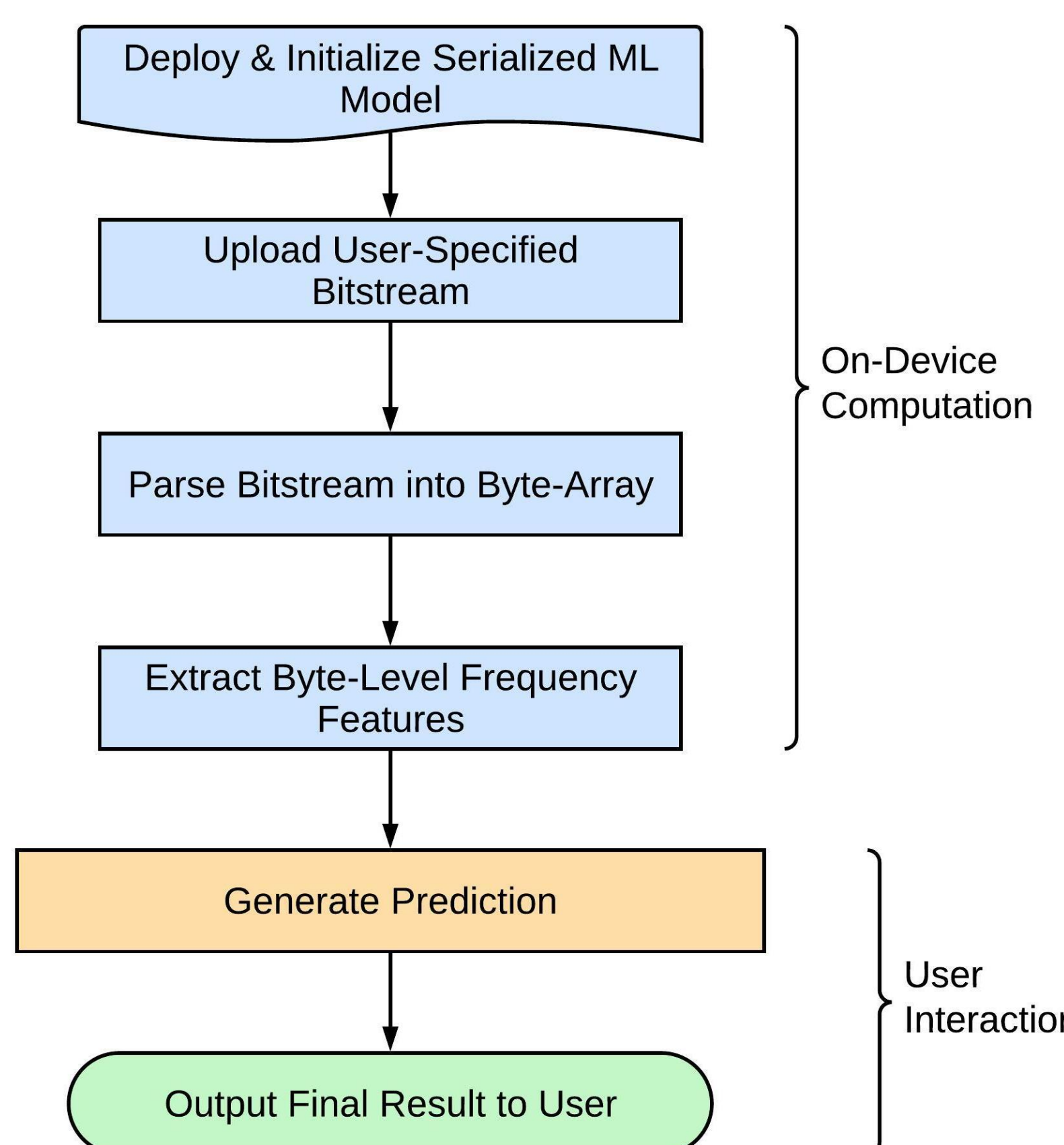


Figure 3. Runtime Prediction Pipeline on PYNQ-Z1 via Jupyter Notebook UI.

References

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Results & Conclusion

Table 1. Performance Metrics (Accuracy, Precision, Recall, F1 Score) for Evaluated Machine Learning Models.

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	0.98 ± 0.02	0.99 ± 0.02	0.98 ± 0.02	0.98 ± 0.02
Gradient Boosting	0.90 ± 0.05	0.88 ± 0.11	0.90 ± 0.06	0.88 ± 0.09
AdaBoost	0.89 ± 0.05	0.87 ± 0.10	0.89 ± 0.05	0.87 ± 0.08
Logistic Regression	0.33 ± 0.07	0.14 ± 0.03	0.36 ± 0.08	0.20 ± 0.05
Naive Bayes	0.95 ± 0.06	0.96 ± 0.05	0.96 ± 0.05	0.96 ± 0.05
SVM (RBF)	0.46 ± 0.10	0.30 ± 0.13	0.51 ± 0.11	0.36 ± 0.12
KNN	0.86 ± 0.04	0.87 ± 0.04	0.86 ± 0.03	0.86 ± 0.04
Decision Tree	0.95 ± 0.00	0.96 ± 0.00	0.95 ± 0.00	0.95 ± 0.00

Table 2. Inference Performance and Classification Results Across Five Deployment Trials on PYNQ-Z1.

Trial	Actual Class	Predicted Class	Load Time (ms)	Feature Extraction Time (ms)	Prediction Time (ms)
1	Malicious RS232	Malicious RS232	239.48	3,157.72	15.03
2	Malicious RS232	Malicious RS232	190.72	3,268.73	16.94
3	Benign AES	Benign AES	23.65	3,167.01	16.81
4	Benign AES	Benign AES	21.85	3,156.55	16.96
5	Malicious AES	Malicious AES	189.02	3,228.77	15.85

- In this work, we proposed an ML-based framework for classifying FPGA bitstreams using static byte-level features and supervised learning. Our pipeline, which leverages dimensionality reduction and a Random Forest classifier, achieved robust multi-class detection performance across both benign and malicious bitstreams.
- On a hold-out test set, **the system reached a TPR of 97.14% and a low FPR of 0.8%**, indicating strong reliability in accurately identifying malicious designs while minimizing false alarms.
- The model was successfully deployed on the PYNQ-Z1 embedded platform with an average latency of 3.35 seconds per classification, demonstrating the feasibility of real-time Trojan detection on resource-constrained systems. Future work will aim to refine the model to handle more evasive threat patterns and further reduce FPR without sacrificing detection performance.

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